

Enhancing Microservices Anomaly Detection via Multimodal Data Fusion in the Wavelet Domain and Spatiotemporal Graph-based Diffusion Probabilistic Model

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Outline

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- 2. Challenges & Contributions
- 3. Methodology
- 4. Experiments
- 5. Conclusion

1. Research Background

Microservices architecture has become increasingly popular in modern software systems, yet its complexity also poses challenges for failure diagnosis. Existing literature has deficiencies in multimodal data mining, mainly in two aspects:

- The mining of multimodal data is insufficient, failing to fully exploit the rich information contained in different types of data.
- The analysis of spatiotemporal features of multimodal data is not thorough, failing to fully explore the potential associations of data in the temporal and spatial dimensions.

1. Research Background

This paper proposes a new method for anomaly detection in microservices. We transform three types of modal data into the wavelet domain and achieve fusion among modalities based on an attention mechanism, fully utilizing the inherent characteristics of multimodal data in the frequency and temporal domains. This fusion process can extract new information that does not exist in the original modalities, thereby enhancing the model's ability to detect anomalies. In addition, we propose a new diffusion probabilistic model (DDPM) based on spatiotemporal graphs, which combines spatiotemporal learning capabilities with the uncertainty measurement of DDPM to generate future samples in a non-autoregressive manner, achieving multi-horizon prediction.

2.1 Challenges

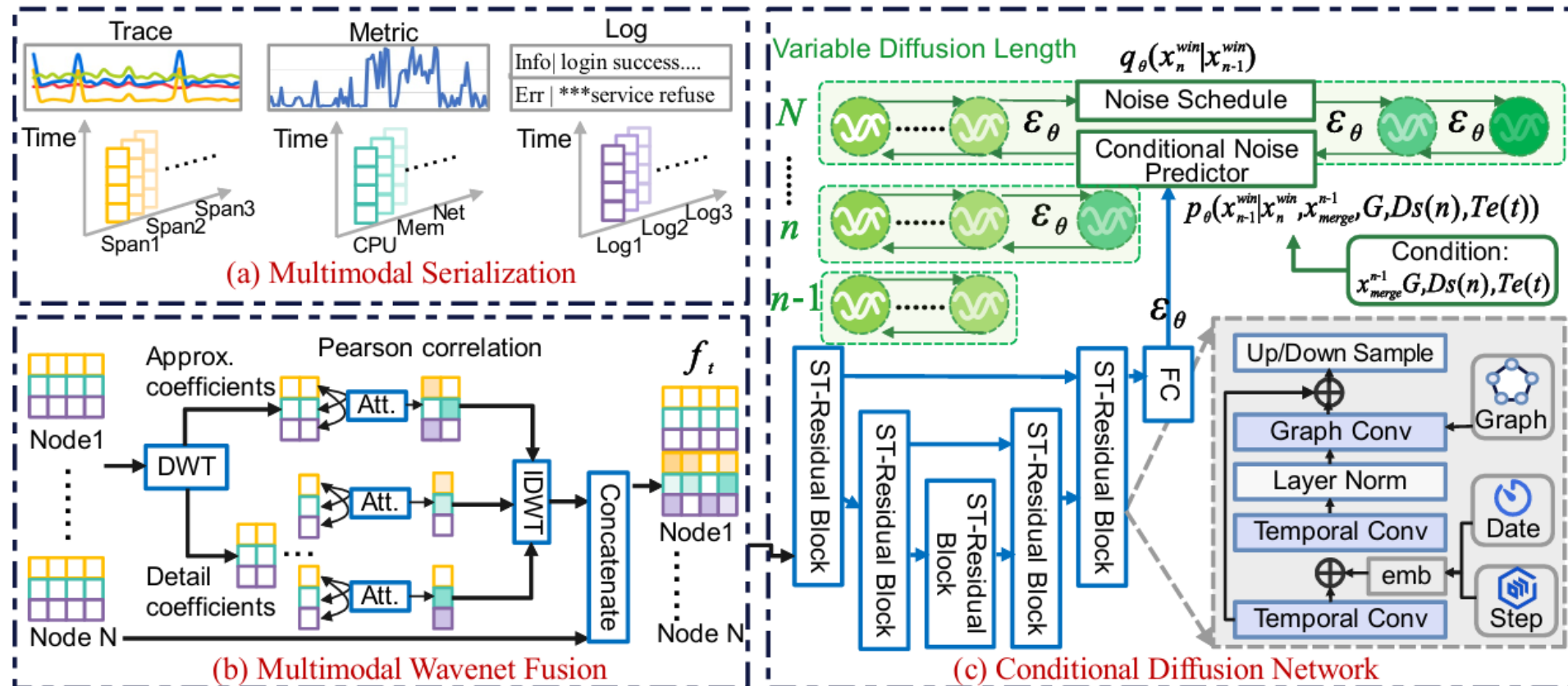
- **Insufficient exploitation on multi-modal data:** Current multimodal anomaly detection approaches inadequately explore cross-modal relationships. Most approaches employ early fusion by simply concatenating multimodal data as multiple channels. More critically, existing research overlooks frequency-domain analysis of multimodal data.
- **Lack of spatiotemporal feature learning:** Accurately identifying anomalous patterns requires modeling both spatial relationships(e.g., service invocation dependencies) and temporal dynamics(e.g., variations over time). However, existing approaches exhibit critical gaps in this regard.

2.2 Contributions

- We are the first to propose feature fusion for multimodal data in the wavelet domain, leveraging a Pearson correlation-based attention mechanism to enhance anomaly detection by combining both temporal and frequency features. We introduce a novel skimming attention mechanism.
- We tackle the problem of probabilistic spatiotemporal anomaly detection from a conditional diffusion perspective with the first shot. FUSION can effectively model the complex spatiotemporal dependencies and intrinsic uncertainties within multimodal data..
- We conduct extensive experiments to demonstrate that FUSION is more effective than state-of-the-art anomaly detectors.

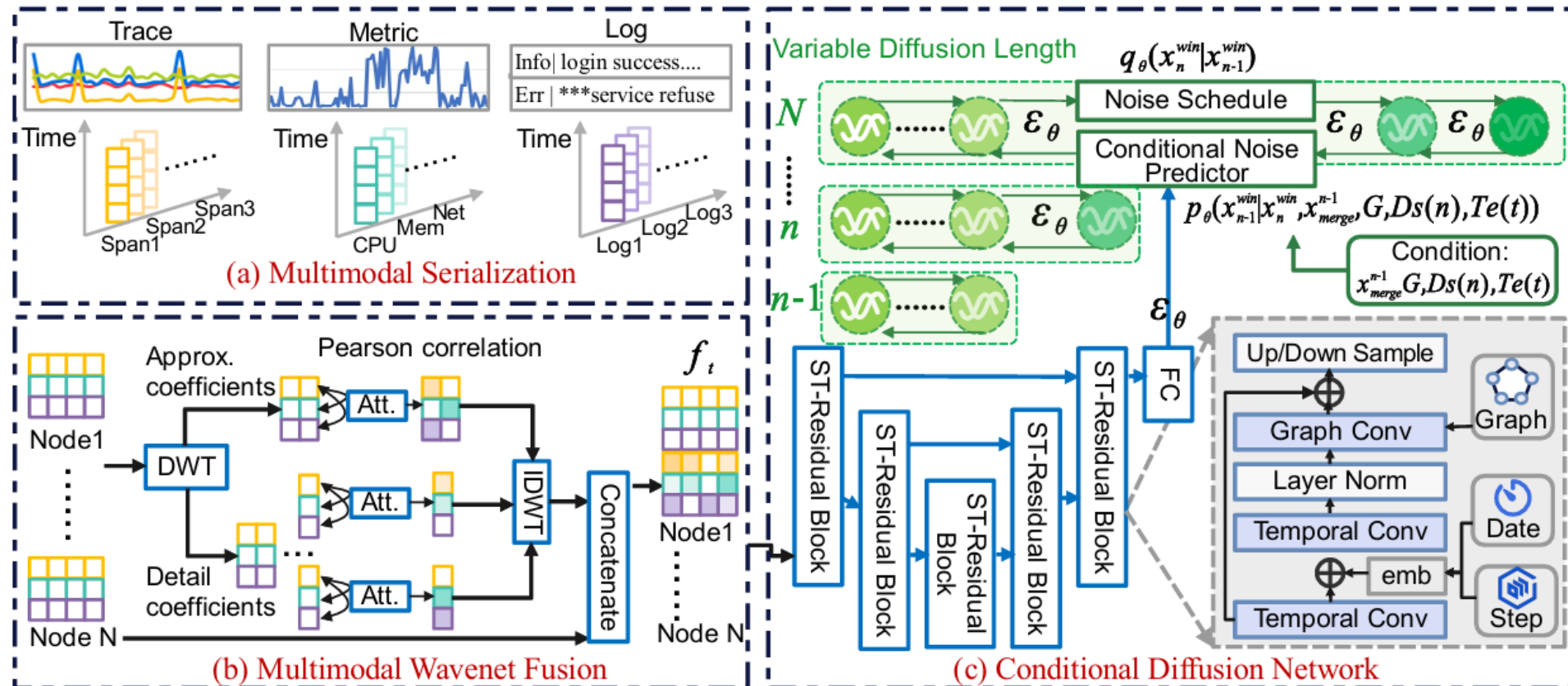
3. FUSION Overview

- Figure presents the overall architecture of FUSION, which consists of multimodal data serialization, multimodal wavelet fusion and conditional diffusion network. First, preprocess the multimodal data.



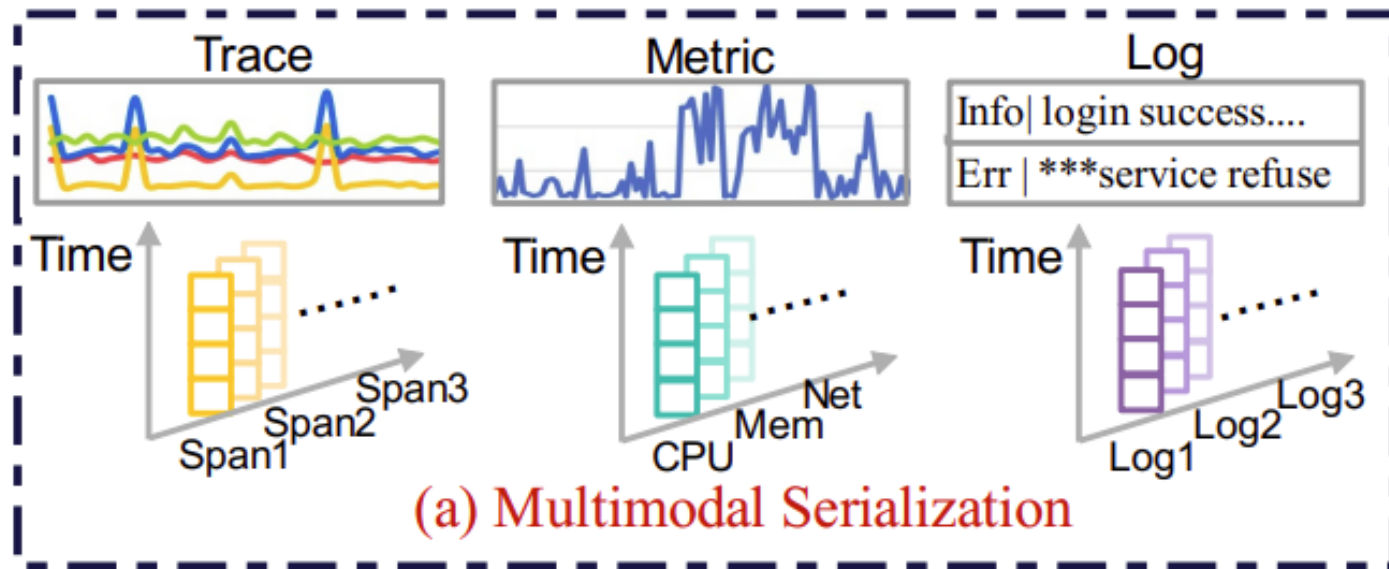
3. FUSION Overview

- Next, these three modalities are combined in the wavelet domain based on an attention-based mechanism. The third stage is the integration of a U-structure-based denoising process during the diffusion phase.



3.1 Multimodal Data Serialization

- **3.1.1 Metric Serialization.** In this paper, we focus on multivariate time-series data, denoted as $\hat{H}_i^M = \{h_{i,0}^M, \dots, h_{i,T}^M\} \in \mathbb{R}^{F_M \times T}$, representing a sequence of timestamped observations of size T for the i -th instance and F_M is the number of metrics.



3.1 Multimodal Data Serialization

- **3.1.2 Log Serialization.** FUSION uses the Drain3 algorithm:
 - (1) Log Template Clustering. We utilize the bert-base-uncased model for embedding vectors and applying the DBSCAN algorithm.
 - (2) Log Entry Serialization. The system calculates the distance between the sentence embedding vectors of log entries and cluster centroids to determine their categories. If a log entry doesn't match a cluster, it's identified as an anomalous template and assigned to a separate category. To align this analysis with system metrics, we count the occurrences of each log category over time.

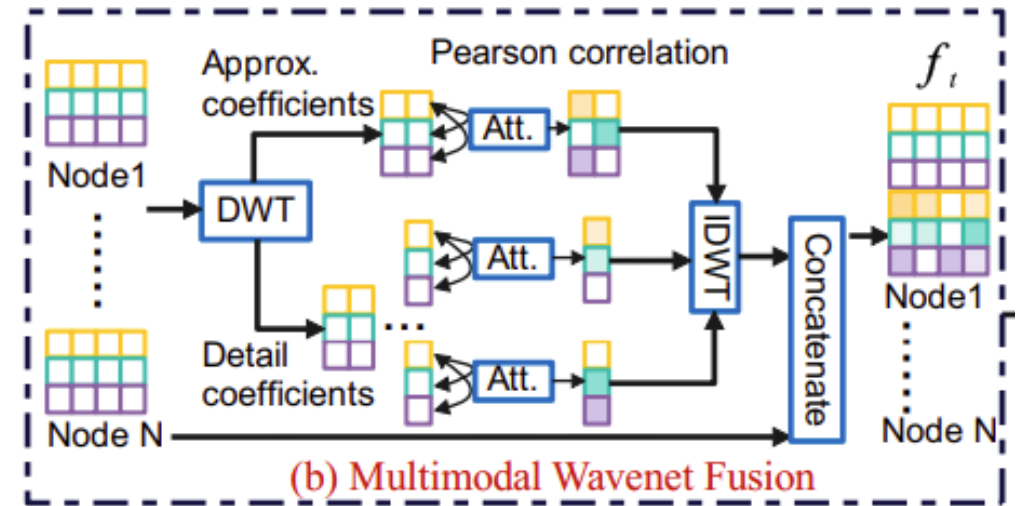
3.1 Multimodal Data Serialization

- **3.1.3 Trace Serialization.** In a microservice system, a trace captures user request executions as spans, which include attributes like trace ID, request type, span ID, parent span ID, service instance ID, start time, and duration. We analyze these spans sequentially to extract structural information and dependencies. Duration is then transformed into a multivariate time series by calculating the average latency, total requests, and status code frequencies per minute.
- **3.1.4 Multimodal Data Construction.** We perform z-score standardization using a sliding historical window on a time series modal. For each instance i at time t , we concatenate the features into a fused feature vector .

3.2 Multimodal Wavelet Fusion

- **3.2.1 Discrete Wavelet Transform.** The DWT module transforms feature vectors into multi-scale frequency representations by breaking down input signals into wavelets.

$$\begin{aligned} \mathbf{cD}_l &= \mathbf{f} * \mathbf{cA}_{l-1} = \sum_{m=1}^{\frac{T_w}{2^{l-1}}} \mathbf{f}[2s - m] \mathbf{cA}_{l-1}[m], \quad s = 1, \dots, \frac{T_w}{2^l} \\ \mathbf{cA}_l &= \mathbf{g} * \mathbf{cA}_{l-1} = \sum_{m=1}^{\frac{T_w}{2^{l-1}}} \mathbf{g}[2s - m] \mathbf{cA}_{l-1}[m], \quad s = 1, \dots, \frac{T_w}{2^l} \end{aligned} \quad (1)$$

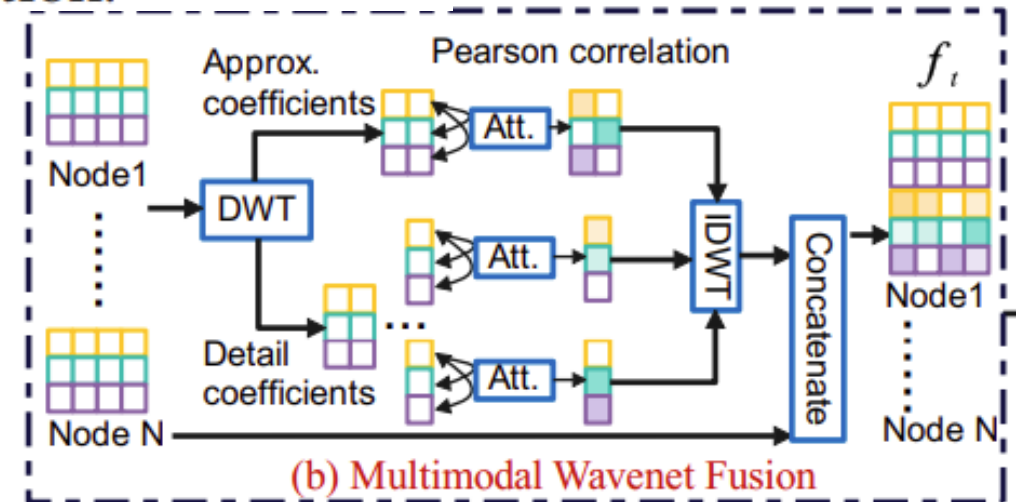


3.2 Multimodal Wavelet Fusion

- **3.2.2 Fusion with Pearson Correlation-Based Attention.** First, we compute the Pearson correlation coefficient matrix between the variables.

$R_{F_i, F_j} = \text{cov}(\mathbf{x}_i^{F_i}, \mathbf{x}_i^{F_j}) / (\sigma_{\mathbf{x}_i^{F_i}} \sigma_{\mathbf{x}_i^{F_j}})$, where $\text{cov}(\mathbf{x}_i^{F_i}, \mathbf{x}_i^{F_j})$ denotes the covariance between variables F_i and F_j of feature matrix $\mathbf{H}_i^{modal}$ for instance i , σ is their standard deviations operation.

$$\alpha_{F_i, F_j, l} = \frac{\exp \left(\text{Att}(\mathbf{cD}_l^{(F_i)}, \mathbf{cD}_l^{(F_j)}) \right)}{\sum_{m=1}^{F'_{modal}} \exp \left(\text{Att}(\mathbf{cD}_l^{(F_i)}, \mathbf{cD}_l^{(m)}) \right)}, \quad (3)$$



3.2 Multimodal Wavelet Fusion

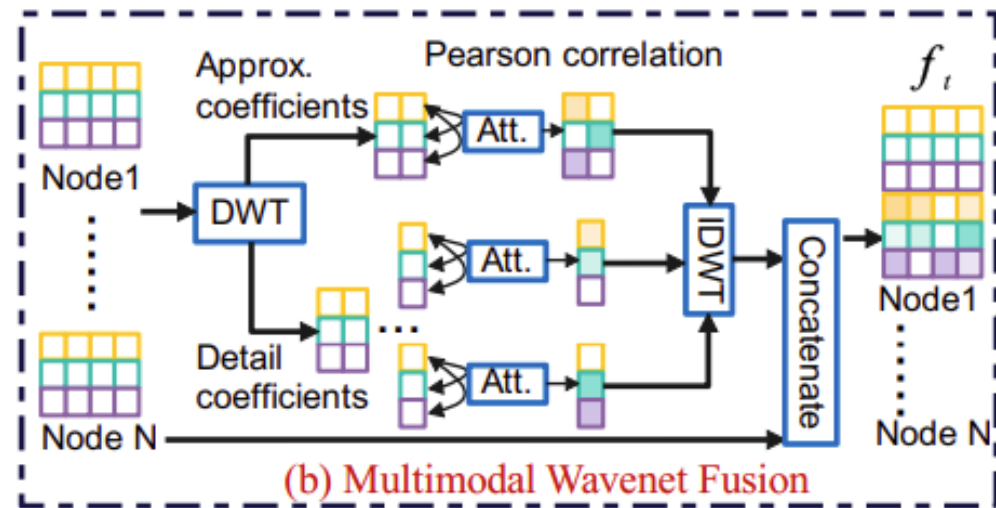
Next, we use a weighted sum to integrate the wavelet coefficients of other variables with itself, computing the context vector $\mathbf{vD}_l^{(F_i)}$ for variable k :

$$\mathbf{vD}_l^{(F_i)} = \mathbf{cD}_l^{(F_i)} + \sum_{m=1}^{F'_{modal}} \alpha_{F_i, F_j, l} \cdot \mathbf{cD}_l^{(m)}. \quad (4)$$

we apply the Inverse Discrete Wavelet Transform (IDWT) to reconstruct their corresponding sequence in the time domain. The process can be formulated as follows:

$$\mathbf{vA}_{l-1} = IDWT(\mathbf{vD}_l, \mathbf{vA}_l) = \mathbf{f}' \star \mathbf{vD}_l + \mathbf{g}' \star \mathbf{vA}_l \quad (6)$$

where $\mathbf{f}'$ and $\mathbf{g}'$ are the synthesis versions of $\mathbf{f}$ and $\mathbf{g}$. Then, $H_i^{merge} = \mathbf{vA}_0$ is the reconstructed time series of the i -th instance.



3.3 Conditional Diffusion Network

- **Problem Formulation:**
 - Spatio-Temporal Graph Modeling: Microservice system represented as a graph $G=(V,A)$.
 - Nodes: service instances.
 - Edges: service dependencies and call relationships.
 - Input and Output
 - Input: historical window $x_h=[x_1, \dots, x_{T_h}]$
 - Output: future window $x_p=[x_{T_h+1}, \dots, x_{T_h+T_p}]$
 - Objective : $p(x_p|x_h,G)$

3.3.1 Conditional Diffusion Model Overview

- **Framework :**
 - A Conditional Denoising Diffusion Probabilistic Model (DDPM).
 - Consists of forward diffusion and reverse denoising processes.
- **Forward Diffusion Process:** Gaussian noise is gradually added to the full time window $x_{\text{win}} = [x_h, x_p]$. Diffusion equation:

$$x_n = \sqrt{\alpha_n}x_0 + \sqrt{1 - \alpha_n}\epsilon$$

- **Reverse Denoising with Conditions:**
 - Reverse Objective: $p_{\theta}(x_{n-1} \mid x_n, x_h, G)$
 - Condition Information: Historical multimodal observations, Microservice dependency graph , Time embeddings and diffusion step embeddings.
 - Prediction Target: The model predicts noise instead of raw values.

3.3.1 Conditional Diffusion Model Overview

- **Training and Inference Procedure**

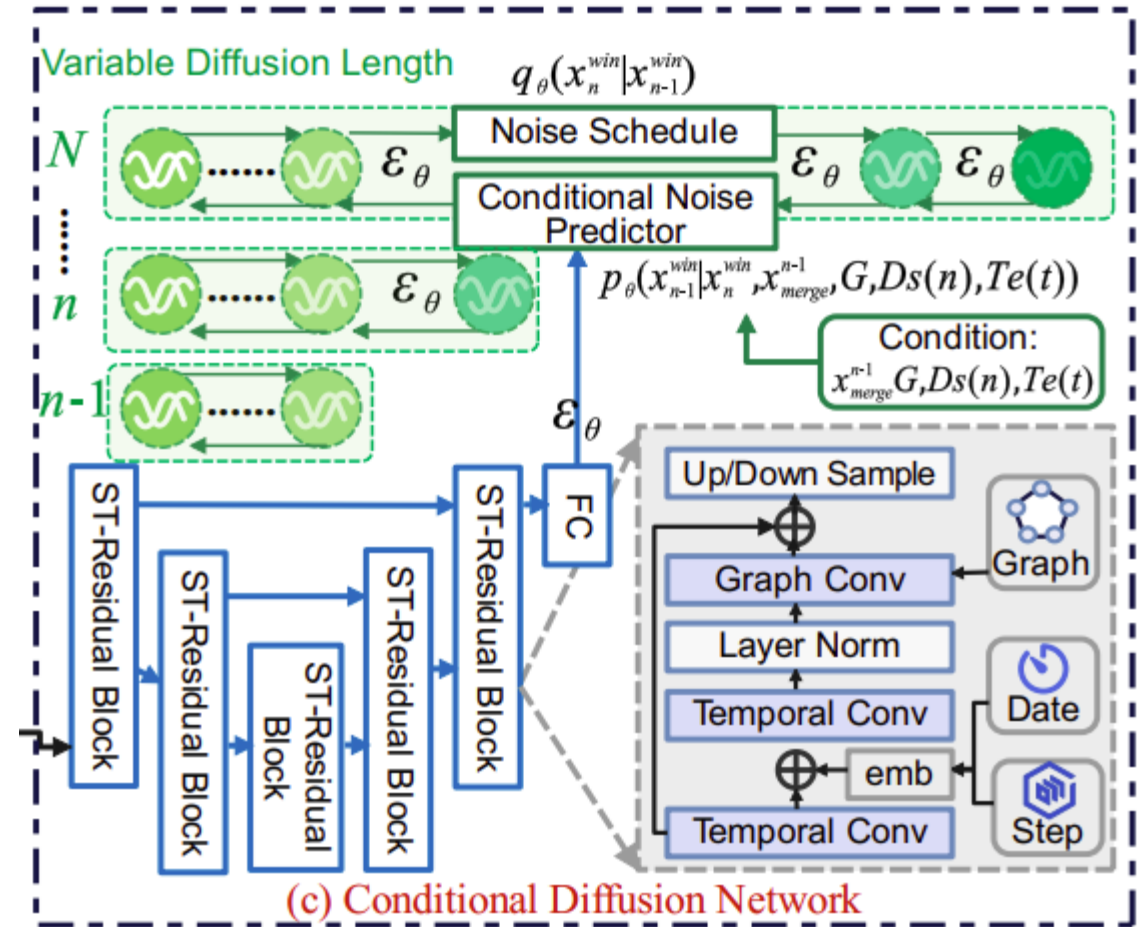
- Training Phase: Uses both historical and future ground-truth data. Optimized via noise prediction loss (MSE).
- Inference Phase: Historical data is fixed as condition. Future steps are initialized with Gaussian noise. Reverse diffusion generates future samples.

- **Future Prediction as a Probability Distribution**

- Sampling Strategy: Multiple diffusion sampling runs with different noise seeds. Each run produces a plausible future trajectory
- Output: Sample ensemble approximates the conditional distribution $p(x_p | x_h, G)$.

3.3.2 Conditional Generation with Unet-based Denoising Network

- **Overall Network Architecture:**
 - U-Net-style architecture with encoder-decoder structure.
 - Multiple Spatio-Temporal Residual Blocks (ST-ResBlocks).
- **Inputs to the Network:**
 - Noisy time window x_{win}^n .
 - Clean historical multimodal data x_{merge}^0 .
 - Microservice dependency graph G .
 - Diffusion step embedding $Ds(n)$.
 - Time embedding $Te(t)$.



3.3.2 ST-ResBlock

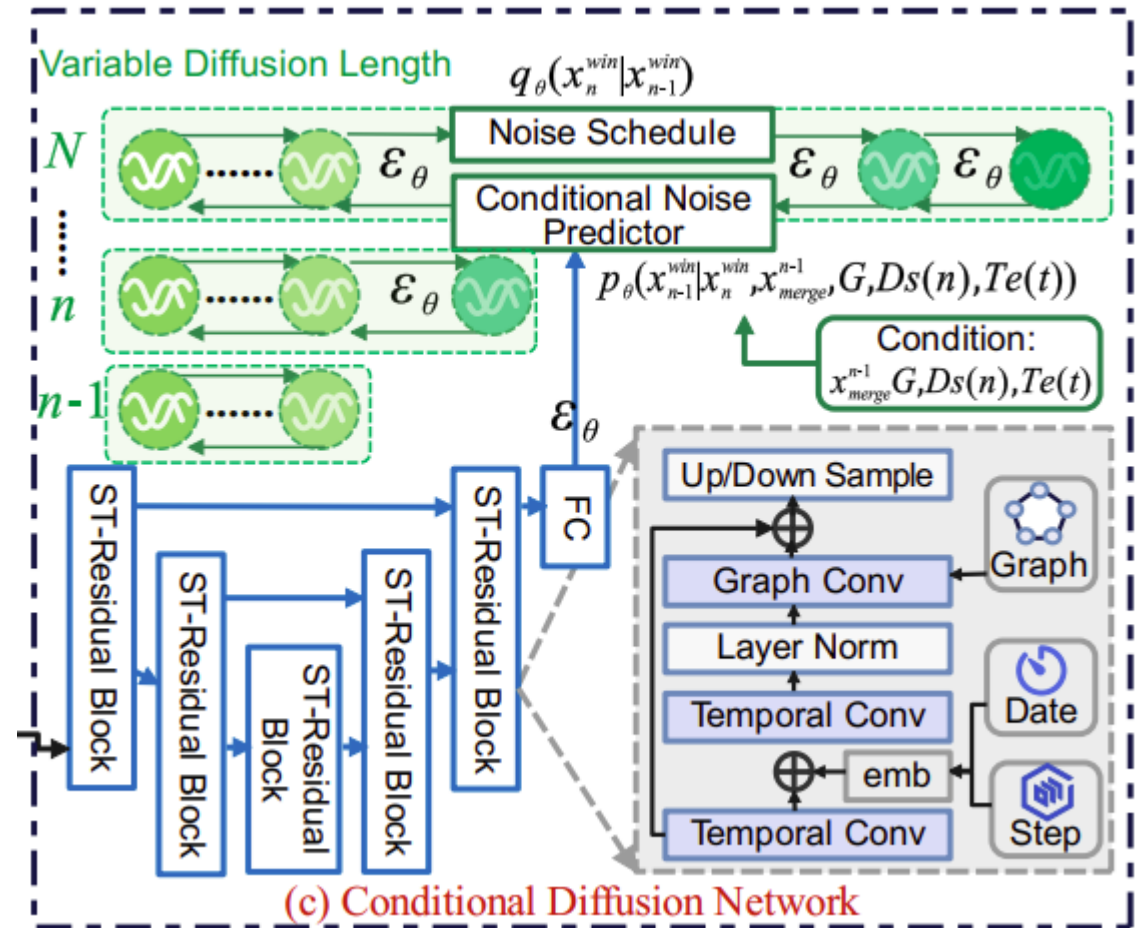
- **ST-ResBlock**:
 - Temporal Convolution Module.
 - Graph Convolution Module.
 - Residual connection with normalization
- **TCN**: Gated convolution mechanism.

$$\Gamma_T(X_k) = P_i \odot \sigma(Q_i) \in \mathbb{R}^{C_{\text{out}}^t \times V \times T},$$

- **GCN**: We use the vanilla GCN.

$$\Gamma_G(\bar{X}_k) = \sigma \left(\Phi \left(A_{\text{gcn}}, \bar{X}_k \right) \mathbf{W}_i \right),$$

$$\Phi_{\text{gcn}}\left(A_{\text{gcn}}, \overline{X}_k\right) = A_{\text{gcn}} \overline{X}_k, \quad \mathbf{A}_{\text{gcn}} = \mathbf{D}^{-1/2}(\mathbf{I} + \mathbf{A})\mathbf{D}^{-1/2}$$



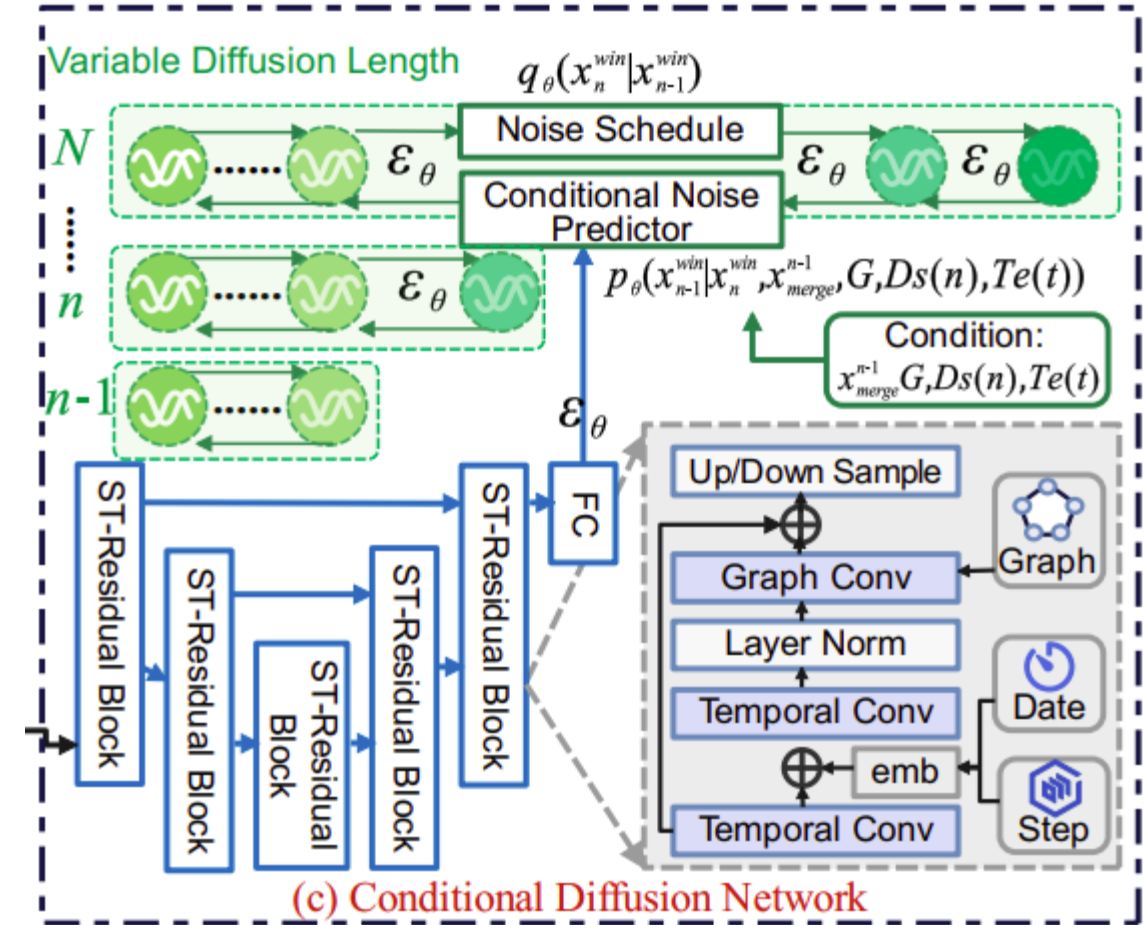
3.3.2 Embedding Injection

- **Diffusion Step Embedding:**
 - Encodes current diffusion step
 - Guides denoising behavior under different noise levels

$$Ds(n) = \left[\dots, \cos\left(\frac{n}{r - \frac{2d}{D}}\right), \sin\left(\frac{n}{r - \frac{2d}{D}}\right), \dots \right]^T$$

- **Time Embedding:**
 - Encodes day, hour, and minute information

$$Te(t) = E_D(t^d) + E_H(t^h) + E_M(t^m)$$



3.3.3 Anomaly Evaluation

- We utilize the Continuous Ranked Probability Score (CRPS) as the abnormal score for each instance during the testing process.
- **The score of each modal data serialization:** $(F, x) = \int_{\mathbb{R}} (F(z) - \mathbb{I}\{x \leq z\})^2 dz$
- $\hat{F}_{sum}(t)$: Sum the (F, x) of three modalities.
- **Anomaly Score:** $AbScore[t] = \mathbb{E}_t \left[CRPS \left(\hat{F}_{sum}(t), \sum_{i=1}^D x_{i,t} \right) \right]$.
- **NDT:** NDT first creates the set e given a set of abnormal scores s of historical observations.

$$e = \mu(s) + z\sigma(s). \quad (13)$$

Then, the threshold Th is calculated by

$$Th = argmax(e) = \frac{(\Delta\mu(s)/\mu(s)) + (\Delta\sigma(s)/\sigma(s))}{|s_a| + |E_{sea}^2|} \quad (14)$$

4.1 Experiment Setup

- **Datasets** : We conducted experimental research on both opensource datasets(AIOps 2022, GAIA, SN) and datasets generated from real-world systems(MMS).

Table 2: Basic information about three datasets.

Data	Failure type	Cases	Percent	Data	Failure type	Cases	Percent
AIOps 2022	File system read	21	12.07%	SN	Network delay	42	29.37%
	File system write	14	8.05%		Network loss	47	32.87%
	Network delay	19	10.92%		CPU stress	54	37.76%
	Network corruption	20	11.49%	MMS	CPU overload	26	9.81%
	Network duplicate	23	13.22%		Memory stress	19	7.17%
	Network drop packet	26	14.94%		High disk	24	9.06%
	CPU load	20	11.49%		Disk partition full	36	13.58%
	Memory load	19	10.92%		I/O latency	23	8.68%
	Container pause	12	6.90%		Database failure	14	5.28%
GAIA	login failure	527	47.95%		Resource exhaustion	38	14.34%
	Memory anomalies	505	45.95%		High packet loss	22	8.30%
	File moving program	36	3.28%		Network partitioning	18	6.79%
	Memory freed label	16	1.46%		Hardware Failures	32	12.08%
	Access denied	15	1.36%		Configuration Errors	13	4.91%

4.1 Experiment Setup

- **Baseline Methods:** We selected nine representative baselines for comparison: TraceVAE , DeepTraLog, Hades , MADMM,MISP, Medicine, AnoFusion, Triple and MSTGAD.
- **Evaluation Metrics:** We choose Precision, Recall, and F1 to evaluate the effectiveness of anomaly methods.
- **Hyperparameters:** We set the batch size as 32 and use the Adam optimizer, we chose the Haar wavelet as the basis function and conducted a three-level decomposition of the time series data, employing high-pass and low-pass filters to produce approximation and detail coefficients. The history length Th and prediction length Tp are set to 100 and 80 for all datasets

4.2 Performance Comparison

- We conducted experiments comparing multiple baseline methods across four datasets.

Table 3: The effectiveness of FUSION in comparison with different methods on three datasets.

Approach	AIOps 2022			GAIA			SN			MMS		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
TraceVAE	0.423	0.406	0.414	0.411	0.396	0.403	0.532	0.511	0.521	0.375	0.321	0.346
DeepTraLog	0.689	0.652	0.670	0.715	0.694	0.704	0.721	0.706	0.713	0.521	0.503	0.512
Hades	0.712	0.694	0.703	0.728	0.719	0.723	0.733	0.724	0.728	0.534	0.521	0.527
MADMM	0.727	0.731	0.729	0.749	0.758	0.753	0.754	0.763	0.758	0.563	0.544	0.553
MISP	0.758	0.736	0.747	0.772	0.761	0.766	0.786	0.767	0.776	0.594	0.547	0.570
AnoFusion	0.783	0.769	0.776	0.794	0.785	0.789	0.784	0.752	0.768	0.694	0.708	0.701
Medicine	0.808	0.793	0.800	0.814	0.797	0.805	0.817	0.804	0.810	0.727	0.711	0.719
Triple	0.843	0.829	0.836	0.854	0.832	0.843	0.918	0.907	0.912	0.827	0.815	0.821
MSTGAD	0.932	0.918	0.925	0.921	0.927	0.924	0.952	0.938	0.945	0.857	0.831	0.844
FUSION w/o Wavelet	0.842	0.834	0.838	0.863	0.873	0.868	0.896	0.872	0.884	0.778	0.763	0.770
FUSION w/o Temporal	0.724	0.713	0.718	0.734	0.727	0.730	0.718	0.704	0.711	0.657	0.621	0.638
FUSION w/o Spatial	0.793	0.779	0.786	0.784	0.768	0.776	0.828	0.813	0.820	0.727	0.735	0.731
FUSION w/o U-structure	0.648	0.618	0.633	0.692	0.682	0.687	0.739	0.716	0.727	0.618	0.603	0.610
FUSION	0.959	0.941	0.950	0.963	0.952	0.957	0.978	0.963	0.970	0.939	0.921	0.930

4.2 Performance Comparison

- We compared the training and detection times of baseline models in Table 4.

Table 4: Comparison of efficiency between FUSION and different methods. Training time (Offline(min)) and average time to diagnose a failure case (Online(s)).

Dataset		TraceVAE	DeepTraLog	Hades	MADMM	MISP	Medicine	AnoFusion	Triple	MSTGAD	FUSION
AIOps 2022	Offline	19.25	77.28	39.00	34.45	112.46	43.91	106.96	47.80	68.32	41.23
	Online	8.41	148.15	10.52	14.38	193.71	15.11	184.91	18.96	21.01	14.22
GAIA	Offline	17.42	68.26	28.91	24.63	109.00	31.45	97.06	32.61	54.98	29.75
	Online	6.51	125.71	5.68	7.03	164.41	10.45	157.60	14.64	17.30	11.61
SN	Offline	5.06	56.57	8.30	14.16	87.49	17.73	82.25	19.83	46.58	20.68
	Online	1.21	118.95	2.38	4.89	134.88	6.59	125.10	8.10	10.23	6.17
MMS	Offline	26.23	89.36	49.47	58.90	148.97	64.12	137.01	69.63	78.67	65.45
	Online	13.45	167.77	37.16	43.05	232.92	59.82	221.56	67.77	85.57	47.22

4.2 Performance Comparison

- We conducted noise analysis.

Table 5: Performance Comparison of FUSION and Baseline Methods under Noise $\sigma = 0.1$ Across Different Modalities.

Approach	Metric_noise		Log_noise		Trace_noise	
	F1	Decrease	F1	Decrease	F1	Decrease
TraceVAE		×		×	<u>0.331</u>	<u>-4.31%</u>
DeepTraLog		×	0.469	-8.37%	0.434	-15.21%
Hades	0.468	-11.27%	0.471	-10.70%		×
MADMM	0.477	-13.80%	0.492	-11.08%		×
MISP	0.493	-13.44%	0.502	-11.86%		×
AnoFusion	0.632	-9.83%	0.634	-9.55%	0.588	-16.11%
Medicine	0.672	-6.53%	<u>0.671</u>	<u>-6.66%</u>	0.639	-11.12%
Triple	0.747	-9.01%	0.623	-24.11%	0.734	-10.59%
MSTGAD	<u>0.809</u>	<u>-4.12%</u>	0.722	-14.43%	0.774	-8.27%
FUSION	0.913	-1.82%	0.879	-5.48%	0.908	-2.36%

Table 6: Performance Comparison of FUSION and Baseline Methods under Noise $\sigma = 0.2$ Across Different Modalities.

Approach	Metric_noise		Log_noise		Trace_noise	
	F1	Decrease	F1	Decrease	F1	Decrease
TraceVAE		×		×	<u>0.328</u>	<u>-5.18%</u>
DeepTraLog		×	0.458	-10.52%	0.412	-19.51%
Hades	0.448	-15.06%	0.462	-12.40%		×
MADMM	0.439	-20.66%	0.483	-12.71%		×
MISP	0.423	-25.73%	0.479	-15.90%		×
AnoFusion	0.583	-16.82%	0.605	-13.69%	0.558	-20.39%
Medicine	0.613	-14.73%	<u>0.645</u>	<u>-10.28%</u>	0.613	-14.73%
Triple	0.689	-16.07%	0.553	-32.64%	0.684	-16.68%
MSTGAD	<u>0.751</u>	<u>-11.00%</u>	0.698	-17.28%	0.745	-11.71%
FUSION	0.892	-4.08%	0.864	-7.09%	0.889	-4.40%

4.3 Ablation Study

- We perform an ablation study using four datasets to assess the impact of each component.

Table 3: The effectiveness of FUSION in comparison with different methods on three datasets.

Approach	AIOps 2022			GAIA			SN			MMS		
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FUSION w/o Spatial	0.793	0.779	0.786	0.784	0.768	0.776	0.828	0.813	0.820	0.727	0.735	0.731
FUSION w/o U-structure	0.648	0.618	0.633	0.692	0.682	0.687	0.739	0.716	0.727	0.618	0.603	0.610
FUSION	0.959	0.941	0.950	0.963	0.952	0.957	0.978	0.963	0.970	0.939	0.921	0.930

5. Conclusion

- We decompose and fuse multimodal data in the wavelet domain. These coefficients are then processed and reconstructed into enhanced feature representations using IDWT.
- We propose a spatiotemporal graph-based Denoising Diffusion Probabilistic Model to capture spatiotemporal features.
- FUSION, utilizing a U-structure architecture, captures multi-scale temporal dependencies and models spatial correlations with Graph Neural Networks. FUSION's non-autoregressive approach efficiently generates multi-horizon predictions, enhancing fault diagnosis in microservices.
- Our experiments show that this approach effectively deals with complex data patterns and noise found in real-world microservice environments